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**“EXAM GUARDIAN: CLOUD-BASED INVIGILATION & CHEATING
DETECTION”**

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of*

CSA1710 – ARTIFICAL INTELLIGENCE

for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

ARTIFICAL INTELLIGENCE AND DATASCIENCE

Submitted by

M.Arshini (192424322)

Under the Supervision of

Dr. B.Nagarajan

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “Exam Guardian: Cloud-Based Invigilation & Cheating Detection” has been carried out by M.Arshini (192424322) under the guidance of Dr. B.NAGARAJAN and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Artificial intelligence and data science program at Saveetha Institute of Medical and Technical Sciences, Chennai.

COURSE COORDINATOR

DR. KRISHNA KUMAR KATHIRESAN

professor

Generative AI

SIMATS Engineering,

SIMATS, Chennai – 602105.

COURSE FACULTY

Dr. Nagarajan B

Professor

Programming

SIMATS Engineering,

SIMATS, Chennai – 602105.

DECLARATION

We, M. Arshini of the [Artificial intelligence and data science], SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “**EXAM GUARDIAN: CLOUD-BASED INVIGILATION & CHEATING DETECTION**” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

Date: September 2026

Signature of the Students

M.Arshini

INDIVIDUAL CONTRIBUTION STATEMENT

The supplied project presentation does not specify a detailed individual task allocation. Therefore, an equal 25% contribution allocation is used below as a documentation assumption. The team should edit the responsibilities if their actual contribution differed.

Student / Register No.	Specific Responsibilities	Design & Development	Testing & Analysis	Report Contribution	Approx. Contribution
M.Arshini / 192424322	System design, development, testing, integration and documentation	Complete project implementation	Complete testing and analysis	Complete report preparation	100%
Total					100%

ABSTRACT

Online examinations have become an important method of assessment in modern education, but maintaining academic integrity during remote examinations remains a significant challenge. Traditional manual invigilation requires continuous human observation and becomes difficult to manage when a large number of students participate simultaneously. The Exam Guardian: Cloud-Based Invigilation & Cheating Detection project proposes an AI-assisted examination monitoring system that combines webcam-based computer vision, object detection, suspicious behaviour identification, evidence capture, and incident logging. The system uses Python and OpenCV to continuously capture and preprocess webcam frames during an examination. The processed frames are passed to the YOLOv5 object detection model to identify selected suspicious events such as mobile-phone presence and multiple persons. The system also considers prolonged head or gaze deviation as a suspicious behaviour through the defined monitoring logic. In addition to webcam monitoring, browser focus monitoring is used to identify tab or window changes during the examination. When a suspicious event is detected, the system captures relevant screenshots and webcam evidence along with timestamps and the reason for the incident. These records are maintained through a structured evidence-storage workflow so that examiners can review suspicious activities after or during the examination. The proposed architecture connects the webcam feed, OpenCV processing module, YOLOv5 classifier, event logger, evidence storage, and examiner-oriented dashboard into a single workflow. The project material reports approximately 90% precision in identifying cheating events under the tested conditions and demonstrates detection of mobile phones, multiple persons, and prolonged head turning. However, performance can be affected by poor lighting and environmental conditions. The system therefore provides a practical prototype for automated examination monitoring while retaining human review for final decisions. Future improvements include larger and more diverse datasets, improved detection under difficult environmental conditions, additional cheating behaviours, stronger privacy and security mechanisms, improved examiner analytics, and production-level cloud deployment.

Keywords: Online Examination, AI Proctoring, YOLOv5, OpenCV, Cheating Detection, Cloud-Based Monitoring

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LIST OF ABBREVIATIONS / SYMBOLS

Abbreviation / Symbol	Description	Unit / Context
AI	Artificial Intelligence	—
CNN	Convolutional Neural Network	—
YOLO	You Only Look Once	Object detection
YOLOv5	YOLO version 5	Object detection model
OpenCV	Open Source Computer Vision Library	Image/video processing
FPS	Frames Per Second	frames/s
RGB	Red, Green, Blue	Image representation
UI	User Interface	Dashboard
PII	Personally Identifiable Information	Privacy context

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Online examinations have become a common method of assessment, particularly with the expansion of remote and hybrid learning. The project presentation identifies fairness and malpractice prevention as major challenges. In a conventional online examination, the candidate may be monitored through a webcam, but continuous human observation is inefficient and difficult to scale. Exam Guardian addresses this engineering problem by combining webcam-based computer vision with event logging and evidence generation.

1.2 Need for the Project

The need for the project arises from the difficulty of detecting and proving suspicious behaviour consistently during online examinations. The supplied project material identifies mobile-phone use, multiple persons assisting a candidate, tab or window switching, and prolonged head/gaze deviation as important cases. A useful engineering solution must detect events quickly, retain evidence, and make the evidence available to an examiner without requiring continuous manual observation.

- Reduce dependence on continuous manual invigilation.
- Detect suspicious objects and people in near real time.
- Capture timestamped evidence for examiner review.
- Provide a structured workflow for incident logging and auditing.
- Support scalable, cloud-oriented examination monitoring.

1.3 Problem Statement

Online examinations are difficult to monitor fairly because candidates can exploit loopholes such as unauthorized mobile-phone use, assistance from additional people, and tab or window switching. Manual monitoring is inefficient, prone to human error and difficult to scale for large examinations. The engineering problem is therefore to develop a real-time system that can capture examination video, identify selected suspicious patterns, generate evidence, log incidents and present the information for examiner review while considering processing latency, privacy and security.

1.4 Project Aim

To design and implement an AI-assisted, cloud-based examination invigilation workflow that detects selected cheating behaviours in real time and stores timestamped evidence for examiner review.

1.5 Project Objectives

1. To capture a candidate's webcam stream continuously during an active examination.
2. To preprocess captured frames and pass them to a YOLOv5-based detection module.
3. To identify selected suspicious events such as mobile-phone presence and multiple persons.
4. To flag prolonged head/gaze deviation and browser tab or window changes.
5. To capture screenshots and webcam evidence with timestamps when incidents occur.
6. To maintain incident logs and provide examiner access through a dashboard-oriented workflow.

7. To evaluate the prototype using detection and validation observations reported in the project material.

1.6 Scope

The project scope includes webcam capture, real-time frame processing, YOLOv5-based detection, suspicious-event classification, evidence capture, timestamped logging, browser-focus monitoring and examiner-oriented review. The project material does not provide a full production deployment specification, a large multimodal dataset, a detailed privacy-compliance audit, or independently reproduced large-scale benchmark results. These are therefore treated as outside the demonstrated scope.

1.7 Expected Engineering Outcome

The intended outcome is a working prototype/workflow for AI-assisted online examination monitoring. The system should accept a webcam stream, process frames, detect selected suspicious events, generate evidence, record incident information and make the resulting records available for review. The project presentation reports approximately 90% precision for identifying cheating events under the tested conditions.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The project presentation provides a literature survey covering computer-vision proctoring, deep-learning-based cheating detection, multimodal cloud proctoring and privacy-aware surveillance. The review is used here as the source basis for identifying common methods, reported benefits and unresolved gaps.

2.2 Review of Existing Work

Title / Work	Author(s)	Year	Methodology / Model	Key Findings / Remarks
Real-Time Face and Gaze Tracking for Online Exam Monitoring	Patel, R.; Singh, K.	2020	OpenCV-based face detection and gaze tracking	Reduced false negatives; detected gaze deviation and multiple faces under stable lighting.
Deep Learning Approach for Cheating Behavior Detection	Chowdhury, A.; Li, H.	2022	CNN-based face/object detection	Reported 89% accuracy for identifying cheating scenarios; performance reduced under poor lighting.
Multi-Modal AI Proctoring System using Cloud Integration	Kumar, P.; Mehta, D.	2023	Video, audio and screen streams on cloud deep-learning pipeline	Improved detection accuracy and scalability; provided administrator alerts.
Privacy-Aware Online Exam Surveillance Framework	Roy, S.; Thomas, B.	2024	Edge-cloud architecture with encrypted evidence storage	Balanced privacy and monitoring accuracy, with some transmission-latency trade-off.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Face / gaze tracking	Simple real-time vision processing	Sensitive to lighting and gaze ambiguity	Supports head/gaze deviation detection.
CNN-based cheating detection	Learns visual cheating patterns	Dataset and lighting dependence	Supports AI classification of suspicious visual events.
Multimodal cloud proctoring	Scalable and combines multiple streams	Higher system and privacy complexity	Motivates cloud-oriented evidence and dashboard workflow.
Privacy-aware edge-cloud	Improved privacy and reduced raw-data transmission	Can increase latency	Motivates controlled evidence handling and security.

2.4 Research / Engineering Gap

- No large-scale multimodal dataset combining video, screen, browser and audio is identified in the project review.
- Existing datasets may have limited participant diversity and scale.
- Webcam-angle coverage of mobile-phone use is weak in many datasets.
- Multi-person assistance and occlusion scenarios are insufficiently represented.
- Frame-level labels may not provide synchronized event logs and timestamps.
- Privacy and consent constraints limit the availability of realistic examination data.

2.5 Proposed Contribution

Exam Guardian combines the practical elements identified in the review into a single workflow: webcam capture, YOLOv5 detection, suspicious-event handling, evidence capture, timestamped logging and examiner review. Its contribution is therefore primarily system integration and prototype implementation rather than the introduction of a new machine-learning architecture. The project focuses on making detected incidents reviewable and traceable instead of relying only on an instantaneous model prediction.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Student / Candidate	Low-latency monitoring with clear examination workflow and minimal unnecessary interruption.
Examiner / Invigilator	Reviewable incidents with reason, timestamp and supporting evidence.
Institution	Fair, scalable and auditable examination monitoring.
System Administrator	Controlled access to stored evidence and reliable incident records.

Functional & Non-Functional Requirements

Requirement	Type	Measurable / Observable Target
Capture webcam frames during the active exam	Functional	Continuous capture while the exam is active.
Detect mobile phone / multiple persons	Functional	Generate an incident when the selected condition is detected.
Monitor tab/window changes	Functional	Capture evidence and terminate the session when the specified event occurs.
Store timestamped evidence	Functional	Evidence associated with incident reason and timestamp.
Real-time operation	Non-Functional	Detection and evidence generation without unacceptable delay during normal test conditions.
Security and controlled access	Non-Functional	Evidence protected through controlled access and encryption as described in the project material.
Reviewability	Non-Functional	Incidents available for examiner auditing.

3.2 Constraints & Applicable Standards

The supplied project material does not identify a formal compliance certification or a specific institutional standard applied during implementation. Therefore, this report does not claim compliance with a named standard. The practical constraints explicitly supported by the project are real-time processing, webcam quality, lighting conditions, privacy/consent limitations, evidence security and scalability.

Constraint / Reference	Requirement	How Applied in This Project
Project privacy requirement	Protect examination evidence and candidate information.	Encryption and controlled access are included in the proposed storage workflow.
Real-time processing constraint	Keep webcam processing responsive.	OpenCV performs continuous frame capture and YOLOv5 processes captured frames.
Lighting / environment constraint	Detection should work under varying conditions.	Project validation reports successful detection in different lighting, while poor lighting remains future work.
No formal standard specified	Avoid unsupported compliance claims.	No certification or standards-compliance claim is made in this report.

3.3 Design Specifications

Parameter	Target / Requirement	Unit / Context	Priority
Webcam capture	Continuous during exam	Video stream	High
Frame preprocessing	Resize and RGB conversion before inference	Per frame	High
Object detection	YOLOv5 inference on captured frames	Per frame	High
Suspicious events	Phone, multiple persons, prolonged head/gaze deviation	Event	High
Evidence	Screenshot + webcam image/overlay + timestamp	Per incident	High
Incident storage	Queryable record with reason and timestamp	Record	High
Detection precision	Project-reported approximately 90%	Precision	High

3.4 Alternative Solutions & Evaluation

Alternative 1: Manual invigilation. Human invigilators observe candidates directly and make decisions based on visible behaviour.

Alternative 2: Rule-based monitoring without AI object detection. Browser focus and simple image-processing rules can trigger alerts, but the visual classification capability is limited.

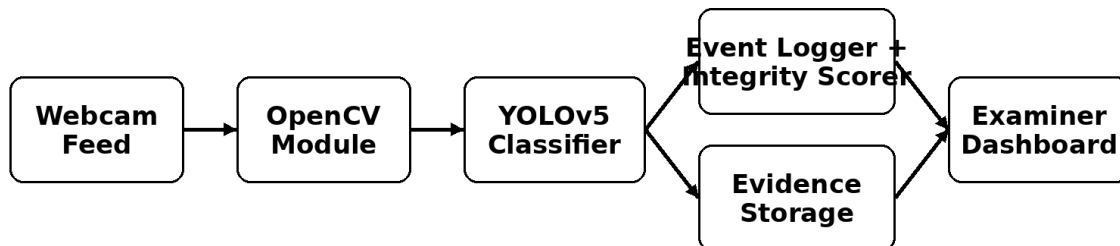
Alternative 3: AI-assisted proctoring using OpenCV and YOLOv5. The selected approach combines real-time capture, visual detection, evidence generation and structured logging.

Criterion	Weight	Alternative 1 Manual	Alternative 2 Rule-based	Alternative 3 AI-assisted
Performance	High	Medium	Medium	High
Cost	Medium	Low–Medium	Low	Medium
Safety / fairness support	High	Medium	Medium	High
Scalability	High	Low	High	High
Evidence generation	High	Variable	Medium	High
Detection flexibility	High	Low	Low–Medium	High

3.5 Selected Design & Detailed Design

The AI-assisted design is selected because it directly addresses the project objectives: real-time webcam processing, object/person detection, evidence capture and examiner-oriented review. The supplied implementation material demonstrates OpenCV capture, YOLOv5 inference, screenshot capture and timestamped logging. The design also leaves browser-focus monitoring as a complementary rule-based input rather than forcing every event through the vision model.

Exam Guardian - System Architecture / Workflow



Real-time webcam processing → AI detection → incident logging/evidence → examiner review

Figure 3.1. Exam Guardian system architecture / workflow.

Systems approach: the webcam feed is captured by OpenCV and passed to the YOLOv5 classifier. Detection results and browser-focus events are converted into incidents by the event-logging workflow. Evidence is stored with timestamps and made available for examiner review through the dashboard-oriented output. This creates a system-level chain from sensing to inference, evidence generation and human verification.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
YOLOv5 object detection	Manual visual observation	Automates selected visual detections	Inference requires compute and can be affected by lighting	Selected because the project requires real-time AI-assisted detection.
Screenshot + webcam evidence	Log-only approach	Provides reviewable proof	Creates additional storage/privacy responsibility	Selected because the project explicitly requires evidence for examiner review.
Cloud-oriented storage/dashboard	Local-only records	Supports centralized review and scalability	Requires secure transmission/storage	Selected to match the cloud-based project objective.
Immediate session termination for tab change	Warning-only workflow	Strong deterrence and clear enforcement	May interrupt legitimate actions or produce false positives	Implemented in project scenario as specified; should be governed by institutional policy in production.

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Use software-based monitoring rather than additional dedicated hardware.	Project relies on webcam, computer and software tools.	Keeps the prototype hardware-light; future cloud scaling must still consider compute/storage use.
Ethics	Monitoring and evidence collection can affect student privacy.	Project review identifies consent/privacy constraints.	Evidence storage and controlled access are required; production use should include institutional privacy procedures.
Health & Safety	Long online examinations and continuous webcam use should avoid unnecessary disruption.	Project emphasizes real-time monitoring and automated termination for specified incidents.	System should minimize false alerts and unnecessary session termination.
Security	Evidence may contain sensitive examination information.	Project specifies encryption and controlled access.	Security is treated as a core storage requirement.

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
False positive detection	Medium	High	High	Use confidence thresholds, event logging and examiner review.	Medium
Poor lighting reduces detection	Medium	Medium	Medium	Improve preprocessing/model training; identified as future work.	Medium
Unauthorized access to evidence	Medium	High	High	Encryption and controlled access.	Low–Medium
Privacy / consent concerns	Medium	High	High	Restrict evidence access and apply institutional privacy procedures.	Medium

Browser focus false trigger	Medium	Medium	Medium	Use robust focus- event logic and clear institutional rules.	Medium
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CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The implementation follows a modular approach. Module 1 captures the webcam feed using OpenCV and prepares frames for inference. Module 2 performs cheating-pattern detection using YOLOv5 and logs suspicious events. Module 3 stores evidence and incident information, while browser-focus monitoring adds a complementary non-vision trigger. The final workflow connects detection, evidence capture and examiner review.

4.2 Software Implementation & Modern Tools Used

Tool / Software	Purpose	Stage Used
Python	Application logic and module integration	All implementation stages
OpenCV	Webcam capture, frame extraction and preprocessing	Module 1
YOLOv5	Object/person detection	Modules 1–2
Browser focus monitoring	Detect tab/window changes	Incident handling
Screenshot / webcam capture	Generate evidence	Incident handling
Timestamped logging / storage	Record incident reason and time	Module 3
Examiner dashboard	Review incident records	Output / review stage

The project presentation includes implementation screenshots showing the code, runtime console and evidence-capture workflow. These screenshots are reproduced below as project evidence.

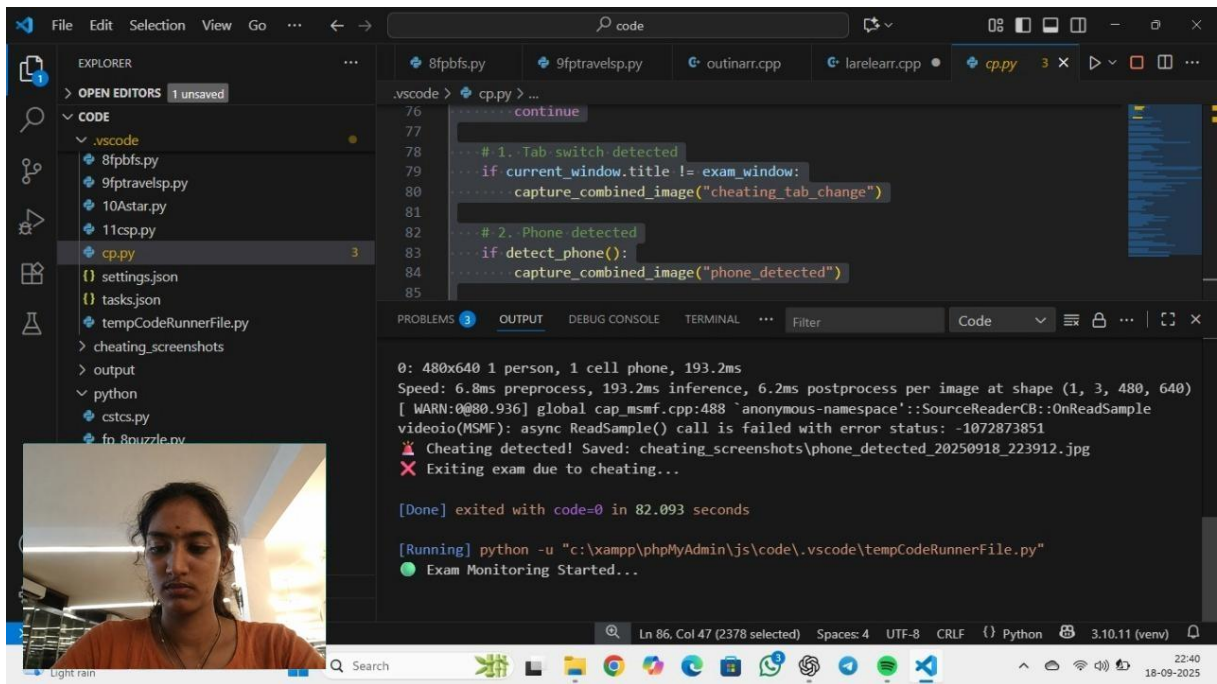


Figure 4.1. Module 1 implementation evidence from the supplied project presentation.

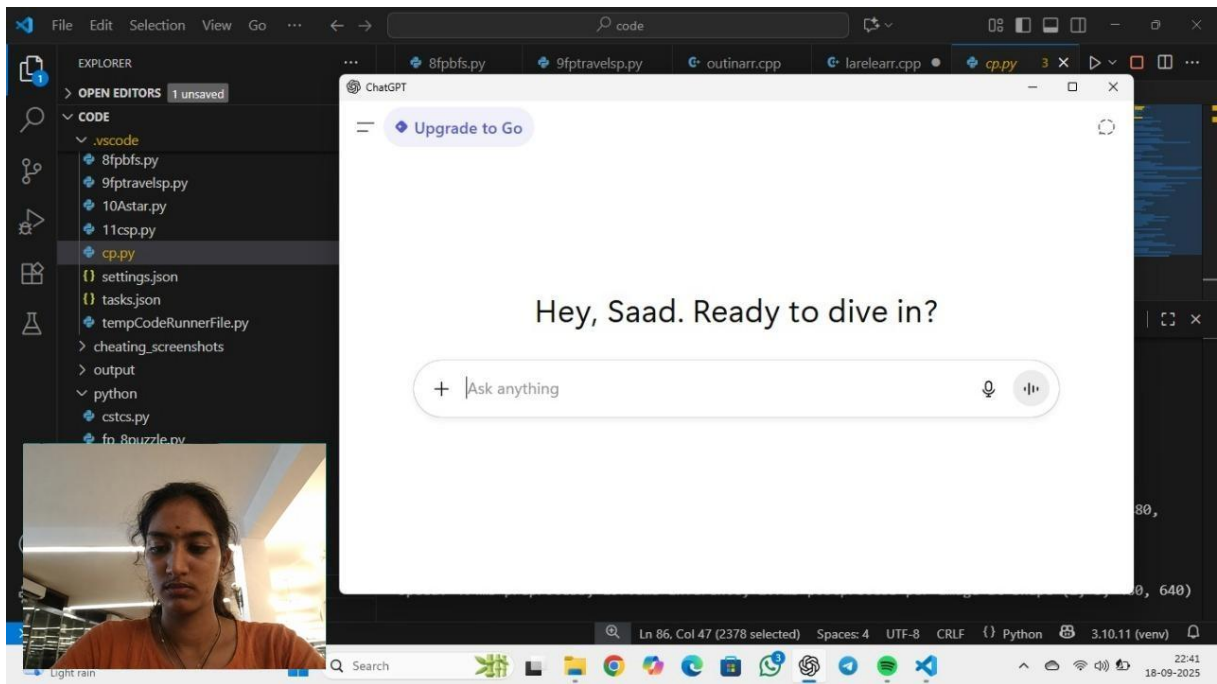


Figure 4.2. Module 1 runtime validation evidence from the supplied project presentation.

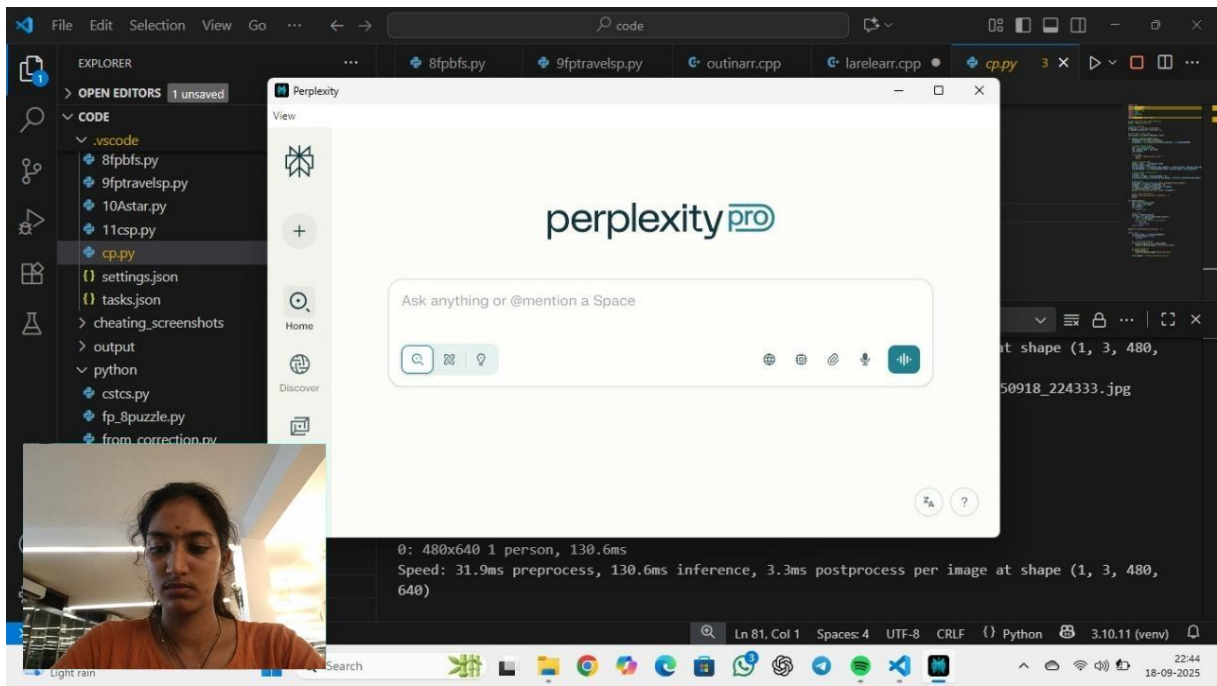


Figure 4.3. Classifier / implementation evidence from the supplied project presentation.

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
Webcam capture	Start examination and observe continuous frame capture.	Frames are captured while the exam is active.
Mobile-phone detection	Place a mobile phone within the webcam view.	Phone detection triggers evidence capture and incident handling.
Multiple-person detection	Introduce more than one visible person.	Additional person is detected and logged as suspicious.
Head/gaze deviation	Maintain head deviation for the specified duration.	Behaviour is flagged as suspicious and logged.
Tab/window change	Change browser focus during the exam.	Desktop screenshot and webcam evidence are captured; project scenario terminates the session.
Evidence logging	Inspect stored incident information after an event.	Reason and timestamp are available for review.

Verification Results

Specification	Required Value / Condition	Achieved Value / Observation	Status
Real-time monitoring	Continuous webcam processing	Implemented through OpenCV loop	Pass
Mobile phone detection	Detect selected phone events	Demonstrated in project scenarios	Pass
Multiple faces	Detect more than one person	Configured and reported as working	Pass
Head turning	Flag prolonged head deviation	Configured and reported as working	Pass
Tab/window change	Capture evidence and terminate session	Implemented in project scenario	Pass
Cheating-event precision	High detection precision	Approximately 90% reported by project	Pass*

*The approximately 90% precision figure is taken from the supplied project presentation and has not been independently reproduced in this report.

4.4 Results

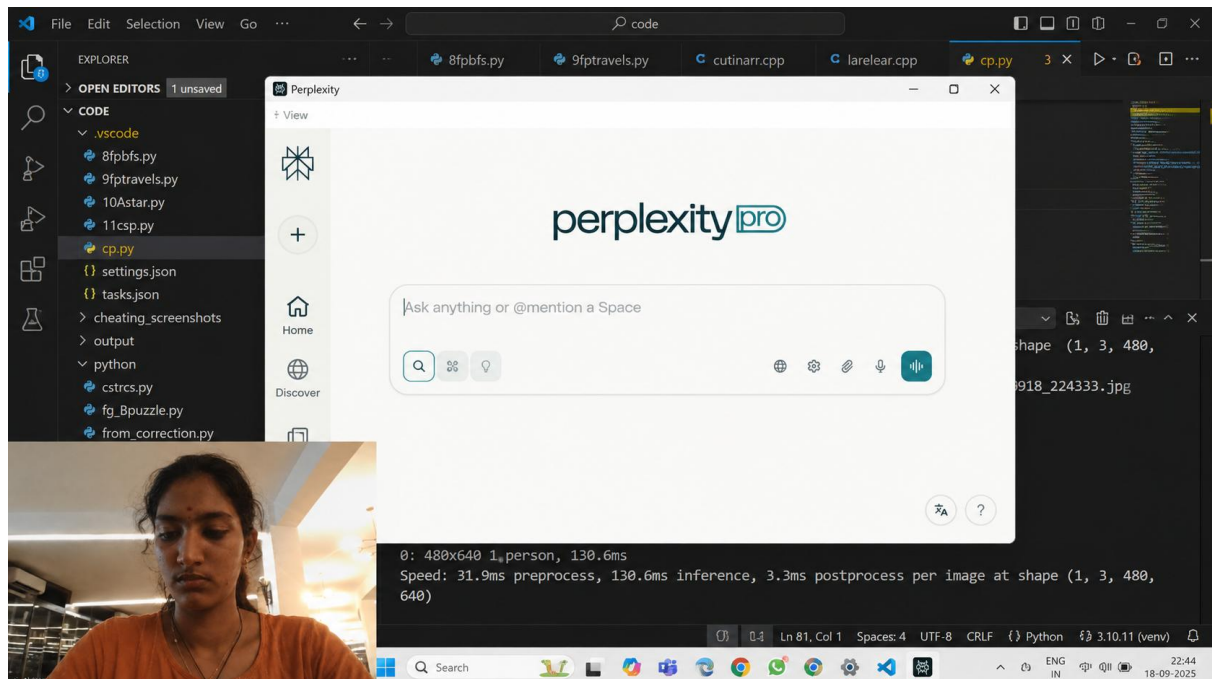


Figure 4.4. Results / evaluation evidence from the supplied project presentation.

The project material reports approximately 90% precision in identifying cheating events. It also states that the integrated system can detect mobile-phone presence, multiple persons and prolonged head

turning, and that detection remained effective across different lighting conditions used during testing. The same material identifies poor lighting and noisy environments as areas for future improvement.

4.5 Data Analysis & Engineering Judgment

The reported precision indicates that the selected AI-assisted approach is capable of identifying a substantial proportion of the targeted cheating events in the project test conditions. However, precision alone does not establish complete proctoring reliability because the supplied material does not provide a full confusion matrix, recall, F1 score, sample count or confidence-interval analysis. Engineering judgment therefore supports using the system as an automated screening and evidence-generation tool with examiner review, rather than treating every model event as an unquestionable final decision.

The observed environmental limitation is important. Computer-vision detection can degrade when the candidate image is poorly illuminated, partially occluded or otherwise outside the conditions represented by the model. The project therefore appropriately identifies improved performance in poor lighting and noisy environments as future work.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Capture webcam stream	OpenCV continuous frame extraction	Fully achieved
Run AI detection	YOLOv5 integrated with webcam stream	Fully achieved
Detect selected cheating behaviours	Phone, multiple persons and head turning are configured	Fully achieved
Capture proof	Screenshots, webcam images and timestamps	Fully achieved
Monitor tab/window change	Browser focus monitoring scenario	Fully achieved
Secure storage / review	Timestamped logging, encryption and controlled access described	Partially achieved / prototype-level evidence
Scalable cloud deployment	Cloud-based architecture objective	Partially achieved; full production deployment evidence is not supplied

4.7 Strengths & Limitations

Strengths:

- Combines real-time webcam processing with AI object detection.
- Produces timestamped evidence rather than relying only on an alert.
- Handles both vision-based events and browser-focus events.
- Provides a structured workflow for examiner review.

- Uses modular components that can be extended in future.

Limitations:

- The supplied results do not include a large-scale benchmark dataset or full statistical evaluation.
- Detection can be affected by poor lighting and environmental conditions.
- The project review identifies limited multimodal and multi-person datasets as an unresolved issue.
- Privacy, consent and secure evidence handling require institutional governance for real deployment.
- The complete production cloud deployment and dashboard implementation details are not fully documented in the supplied material.

4.8 Project Planning, Roles & Budget

The supplied project material does not contain a dated Gantt chart or a detailed expenditure sheet. To preserve traceability, the planning information below is presented as a project-structure summary rather than as claimed historical spending.

Stage	Planned / Implemented Activity	Primary Output
1	Problem understanding and literature review	Requirements and gap analysis
2	Webcam capture module	Real-time video stream
3	AI detection module	YOLOv5-based event detection
4	Evidence and storage module	Screenshots, timestamps and logs
5	Scenario testing	Phone, multiple-person and tab-change validation
6	Integration and documentation	Final workflow and report

Student	Assigned Responsibility	Actual Contribution
M.Arshini	System implementation / integration	Team contribution; exact allocation to be confirmed
	AI detection / classifier work	Team contribution; exact allocation to be confirmed
	Storage / testing / integration support	Team contribution; exact allocation to be confirmed

Item	Estimated Cost	Actual Cost
Existing laptop / computer	Existing resource	Not specified
Existing webcam	Existing resource	Not specified
Python / OpenCV / YOLOv5 software	Open-source / project software	Not specified
Cloud/storage resources	Prototype-dependent	Not specified

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

Exam Guardian addresses the engineering challenge of monitoring online examinations through an AI-assisted, cloud-oriented workflow. The system combines OpenCV webcam capture, YOLOv5 object/person detection, suspicious-behaviour handling, browser-focus monitoring, evidence capture and timestamped incident storage. The supplied project material reports approximately 90% precision for identifying cheating events and demonstrates detection scenarios involving mobile phones, multiple persons and prolonged head turning. The project therefore provides a practical prototype for automated screening and evidence generation while retaining examiner review as an important part of the decision process.

5.2 Future Work

- Improve detection performance in poor lighting and noisy environments.
- Expand the set of detectable cheating behaviours.
- Develop richer examiner reports and analytics.
- Support additional online examination platforms.
- Build larger and more diverse multimodal datasets.
- Improve privacy-aware processing and secure evidence management.
- Evaluate recall, F1 score, false-positive rate and latency using a larger controlled test set.
- Complete production-grade cloud deployment and scalability testing.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Real-time webcam acquisition and preprocessing using OpenCV.
- Practical use of YOLOv5 for object/person detection.
- Integration of visual detection with event logging and evidence capture.
- Design considerations for privacy, evidence security and examiner review.

Engineering & Professional Skills Developed

- System-level decomposition into capture, detection and storage modules.
- Prototype integration and testing.
- Technical documentation and presentation.
- Risk identification and consideration of human review in AI-assisted decisions.

Individual Reflection – M.Arshini (192424322)

The main engineering challenge was integrating real-time webcam capture, AI-based detection and evidence handling into one coherent examination-monitoring workflow. A key engineering decision was to use OpenCV with YOLOv5 because the project requirements call for continuous video processing and detection of selected suspicious events. The project strengthened my understanding of computer-vision integration, incident logging, testing and security considerations for examination evidence. If the project were repeated, I would expand the test dataset, evaluate recall and false-positive rates in addition to precision, and strengthen the privacy and cloud-deployment design.

REFERENCES

The following references are presented in a consistent IEEE-style format. The four literature entries below are the works listed in the supplied project presentation. Full bibliographic details such as journal, volume, issue, pages and DOI were not supplied, so they are not fabricated here.

- [1] R. Patel and K. Singh, “Real-Time Face and Gaze Tracking for Online Exam Monitoring,” 2020.
- [2] A. Chowdhury and H. Li, “Deep Learning Approach for Cheating Behavior Detection,” 2022.
- [3] P. Kumar and D. Mehta, “Multi-Modal AI Proctoring System using Cloud Integration,” 2023.
- [4] S. Roy and B. Thomas, “Privacy-Aware Online Exam Surveillance Framework,” 2024.
- [5] OpenCV, “Open Source Computer Vision Library,” software resource used for webcam capture, frame extraction and image processing.
- [6] YOLOv5, object-detection framework used in the project for visual detection.
- [7] Python, programming language used for project logic and integration.

APPENDICES

Appendix A – Detailed Algorithms / Essential Technical Logic

A.1 Webcam Feed Capture

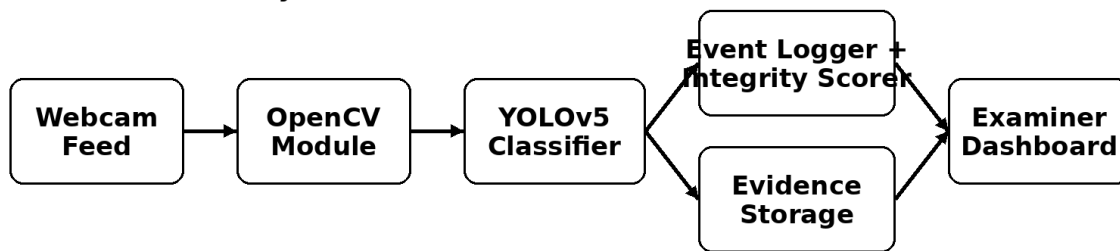
8. Initialize the webcam using OpenCV.
9. While the examination is active, capture a frame.
10. Preprocess the frame by resizing and converting it to RGB.
11. Pass the frame to the YOLOv5 detection model.
12. If a selected suspicious object/person is detected, capture evidence.
13. Save the evidence and timestamped event record.
14. Continue processing until the examination ends.
15. Release the webcam.

A.2 Cheating Pattern Classifier

16. Receive frames from the webcam module.
17. Preprocess the frame.
18. Run YOLOv5 inference.
19. Check for mobile-phone or multiple-person detections.
20. Capture evidence and trigger an alert when required.
21. Check head/gaze deviation rules and log suspicious behaviour.
22. Continue until the examination ends.

Appendix B – Engineering Drawings / Schematics

Exam Guardian - System Architecture / Workflow



Real-time webcam processing → AI detection → incident logging/evidence → examiner review

Appendix Figure B.1. System-level architecture derived from the workflow in the project presentation.

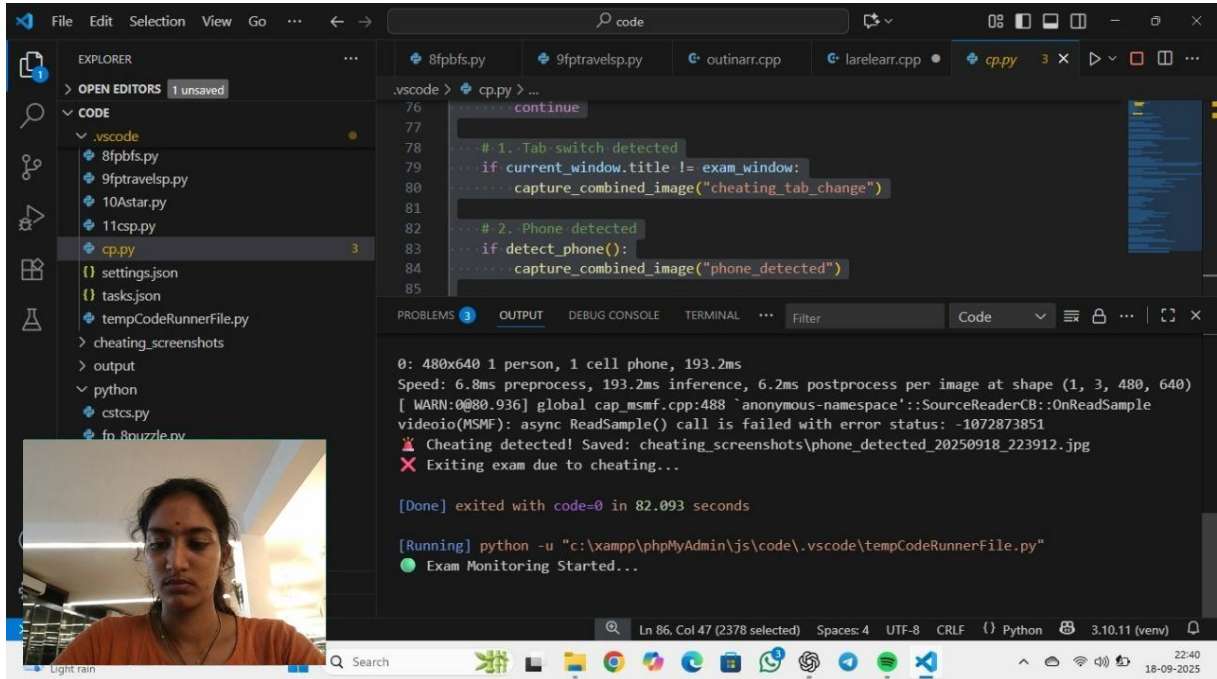
Appendix C – Source Code / Code Repository Information

The supplied project presentation contains screenshots of the implementation and runtime environment. The complete source repository path is not provided in the source material. Therefore, this report documents the essential algorithms rather than inventing a repository URL.

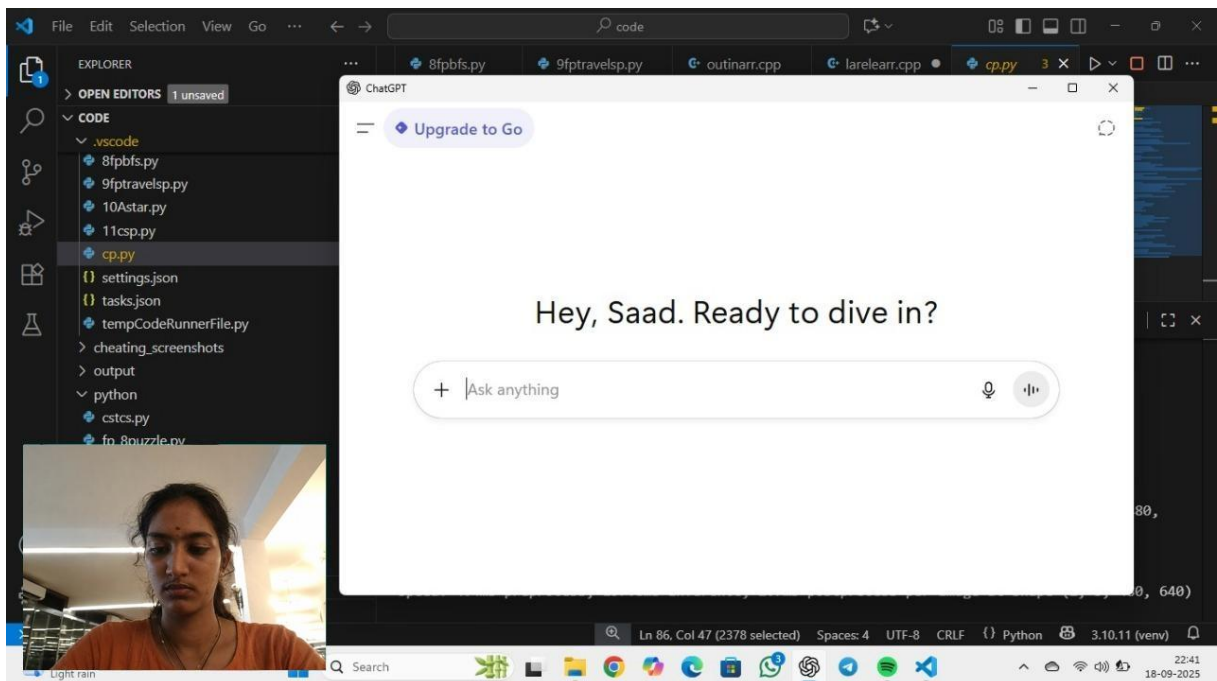
Appendix D – Experimental / Raw Data

The supplied material reports an approximately 90% precision result and qualitative validation across selected scenarios. A complete raw dataset, confusion matrix, per-test counts and statistical analysis are not included in the supplied material.

Appendix E – Project Evidence Screenshots



Appendix E.1. Webcam / detection implementation evidence.



Appendix E.2. Runtime evidence reproduced from the supplied presentation.

Appendix F – Ethics / Institutional Approval

No separate institutional approval document is included in the supplied project material. Because the system processes webcam images and examination evidence, any production deployment should follow the applicable institutional privacy, consent, data-retention and access-control requirements.

Appendix G – Individual Contribution Evidence

The contribution table in the front matter records a provisional equal contribution allocation because the supplied project material does not provide individual task percentages. The team should update this section if official contribution records are available.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

The following mapping follows the structure of the supplied capstone-report template and identifies where evidence is located in this report.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3.7
Teamwork and leadership	SO5	Ch. 4.8
Experimentation, analysis, engineering judgment	SO6	Ch. 4.3–4.6
Independent acquisition of new knowledge	SO7	Ch. 5.3
Standards / applicable constraints	SO2	Ch. 3.2
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3.7
Risk assessment and mitigation	SO2/SO4	Ch. 3.7
Security considerations	Relevant SO	Ch. 3.7
Integrated/system-level engineering	SO1/SO2	Ch. 3.5
Project planning and management	SO5	Ch. 4.8
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4.8